Data Administration in Information Systems

Project

This project is based on real-world data provided by Gov.UK. It is a statistical data set named Price Paid Data. It contains on all property sales in England and Wales that are sold for value and are lodged with HM Land registry. For more information about where the data came from, and their contents, please refer to the Price Paid Data portal. The corresponding metadata (i.e., names and descriptions of the columns is available here)

In particular, we use two datasets from Price Paid Data:

- 2025 csv data
- 2024 csv data

Tasks

In this project, you are asked to perform the following tasks using SQL Server 2022:

1. (Database creation and data insertion) Create tables in SQL Server with names `PricePaidData2025` and `PricePaidData2024` load the 2025 and 2024 data contained in CSV files indicated above into tables named `ShortPricePaidData2025` and `ShortPricePaidData2024`. The columns of these tables should be the following ones (i.e., there are some columns from the original data sets that you do not need to consider): `Transaction unique identifier`, `Price`, `Date of Transfer`, `Postcode`, `Property Type`, `Old_New`, `PAON`, `Street`, `Locality`, `Town`, `District`, `County`. Present the SQL instructions used to create the tables and the number of tuples contained in each table.

2. (Data storage) Using the `ShortPricePaidData2025` table, implement a partitioning scheme such that records are stored in separate partitions based on the `county` attribute (i.e., records referring to the same county are stored in the same partition). Present and execute the necessary SQL instructions to create the partitioned table structure and load the 2025 Price Paid data into the appropriate county-level partitions. Finally, write an SQL query to report the number of records contained in each partition and execute it. Present both the query and the result obtained.

3. (Using indexes and estimating query plan cost) For this task, restore the database to its state at the end of Task 1. The following query returns the average price in 2025 of new properties by locality in London (i.e., town = London):

```
SELECT locality, AVG(price) AS average_new_property_price
FROM ShortPricePaidData2025
WHERE town = 'London'
AND Old_New = 'Y'
GROUP BY locality
ORDER BY average_new_property_price DESC;
```

   Perform a study of the Estimated Subtree Cost for various possible indexes that may be useful to optimize this query. Present the execution plan and the Estimated Subtree Cost obtained with each index. Compare this with the execution plan before creating indexes. Also, include the SQL instructions needed to create and to drop each index.

4. (Using indexes and estimating query plan cost) For this task, restore the database to its state at the end of Task 1. The following query returns districts with increased average prices from 2024 to 2025.

```
SELECT p24.district,
       p24.avg_price AS avg_2024,
       p25.avg_price AS avg_2025
FROM ( SELECT district, AVG(CAST(price AS BIGINT)) AS avg_price
       FROM PricePaidData2024
       GROUP BY district
      ) p24
JOIN ( SELECT district, AVG(CAST(price AS BIGINT)) AS avg_price
       FROM PricePaidData2025
       GROUP BY district
      ) p25
ON p24.district = p25.district
WHERE p25.avg_price > p24.avg_price;
```

Perform a study of the Estimated Subtree Cost for every possible join algorithm and group-by algorithm that this query could use in SQL Server. Present the SQL instructions, the execution plan, and the Estimated Subtree Cost obtained for each join and group-by algorithm. How do you explain the results obtained?

5. (Query tuning) Consider the following SQL query that returns the number of properties that appear in both `ShortPricePaidData2024` and `ShortPricePaidData2025`, per county:

```
WITH p24 AS (
   SELECT DISTINCT county, PAON, street, city
   FROM PricePaidData2024
),
p25 AS (
   SELECT DISTINCT county, PAON, street, city
   FROM PricePaidData2025
)
SELECT p24.county, COUNT(*) AS properties_sold_both_years
FROM p24 JOIN p25
   ON p24.county = p25.county
   AND p24.PAON = p25.PAON
   AND p24.street = p25.street
   AND p24.city = p25.city
GROUP BY p24.county;
```

Propose a query rewriting strategy, using a single query, to increase the performance of the initial query. Justify and then present the execution plans and the Estimated Subtree costs for both, the original and the rewritten query.

6. (Transactions and Isolation levels) Consider a variant of the query presented in 3., without filtering the town value nor the type of property. This is a typical analytical query that computes aggregate statistics (in this case, average price by locality). Suppose an environment where concurrent users execute different transactions: one of them is the analytical query and another one is a query that inserts new records in table `ShortPricePaidData2025`. Demonstrate how different SQL transaction isolation levels can affect the correctness and consistency of these analytical queries. In particular, show how an appropriate isolation level can be used to prevent

dirty reads and non-repeatable reads. Hint: use the SQL instruction `SET TRANSACTION ISOLATION LEVEL`.

Report

Prepare a report to present and describe all the SQL code, queries, execution plans and results for each task. There is no specific template, and any text processor will do, as long as it has the capability of generating an A4-sized PDF document for submission on Fénix.

As you present the results (for example, an Estimated Subtree Cost for a certain query), you are welcome to comment on why that result is better or worse than other options. Regarding the execution plan, when the task asks for it, present the screen capture. Your comments should reveal a good knowledge of the topics addressed in this course.

Carrying out the tasks in this project is only part of the goal. Understanding what is going on is the ultimate goal. If needed, you are encouraged to do some research on your own. Cite the sources you used in a section named “Bibliography” at the end of the report. Furthermore, if you end up using AI tools, please cite them in a section named “Declaration” before Bibliography and describe for which purpose you used them.